

Formal Verification of TFPT Carrier Rigidity in Lean 4

An Audited Dependency Graph from the Boundary Involution to the Standard-Model $3 + 2$ Hypercharge Carrier

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Abstract

This note documents a Lean 4 formalisation of the algebraic carrier rigidity theorem of TFPT Paper 2 (*Carrier Rigidity, Hypercharge, and the Standard-Model Packet from Boundary Polarization*), up to and including the determinant-normalised $3 + 2$ hypercharge carrier $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$. The bundled top-level theorem consumes four explicit interfaces: a carrier involution, Higgs and Yukawa rank witnesses, and a primitive oriented determinant-preserving carrier condition. Each interface is refinable upstream via a separately formalised Lean theorem. In v10 each upstream interface is closed by a further structural layer: the Calderón idempotent $\pi^2 = \pi$ is itself produced from a `BoundaryPolarization` ($H = H_+ \oplus H_-$); the Higgs algebraic shadow is identified with the degree-1 piece of the bivariate polynomial ring $K[X_0, X_1]_1 = \text{MvPolynomial.homogeneousSubmodule (Fin 2) K 1}$ (the algebraic-geometry "Proj-side" presentation of $H^0(\mathbb{P}^1, \mathcal{O}(1))$); and the Stage D Yukawa trilinear form is constructed explicitly from any basis of a rank-3 space, with injective and surjective contraction proved by direct basis computation. The Yukawa Stage D characterisation $\exists \omega \text{ (CI} \wedge \text{CS)} \iff \dim E = 3$ now holds under `Nontrivial E`. The unique integer pair $(q_-, q_+) = (-2, 3)$ follows from a general (m, n) lattice-rigidity theorem applied to the structurally bundled `PrimitiveOrientedDeterminantCarrier 3 2` input. Lean proves the unique $3 + 2$ hypercharge carrier, trace zero, and the carrier polynomial $6Y^2 - Y - 1 = 0$. The project builds against Mathlib v4.29.1 with zero sorry, zero admit, no domain-specific axioms, no unsafe/opaque/partial escape hatches, and every load-bearing theorem depends only on the three standard Lean axioms (`propext`, `Classical.choice`, `Quot.sound`). A seven-check `audit.sh` gate is shipped; the headline-theorem checks combine Lean `#check` elaboration (`AuditCheck.lean`) with strict `example` : `<exact type> := <theorem>` signature locks (`AuditContract.lean`), so neither silent renames nor silent weakening of theorem statements can mask regressions.

Scope box: inputs, contribution, non-claims, audit surface

Inputs from previous papers. Paper 1 (boundary primitive kernel) for the *conceptual* derivation of the carrier involution from the Calderón polarisation. The Lean project consumes only algebraic data: an involution element ε with $\varepsilon^2 = 1$, plus Stage-A certificates for the Higgs and Yukawa rank discharges.

New contribution. A reproducible Lean 4 project `TfptCarrier` (Mathlib v4.29.1) that elevates four previously imported inputs to formal Lean theorems:

- boundary involution $\Rightarrow$ orthogonal idempotents;
- general integer rigidity $\Rightarrow$ unique primitive pair;
- structural trace identity $\Rightarrow \text{Tr } Y = 0$ from rank data alone;
- determinant character of the carrier torus $\Rightarrow$ Diophantine condition $m q_- + n q_+ = 0$ as

a corollary (universal- λ form, with a single- point helper at $\lambda = 2$).

Two further inputs ($\dim E_+ = 2$ and $\dim E_- = 3$) are represented as structured hypotheses, not bare numerical assumptions. They mark the exact interfaces where the future geometric (Higgs) and representation-theoretic (Yukawa) proofs must enter.

For the Yukawa case, this note now provides *four* interface layers, with the upstream stage carrying genuine algebraic content:

- *Stage A* (YukawaRankCertificate) — a bare linear equivalence $E_- \simeq K^3$. Used by the bundled `CarrierData.CarrierPremises` as a rank witness; refinable from any of the upstream stages.
- *Stage B* (YukawaTopForm) — a linear equivalence $\Lambda^3 E_- \simeq K$ (an algebraic top form on E_-).
- *Stage C* (PrimitiveIndecomposableYukawaCoupling) — a linear equivalence $E_- \simeq (\Lambda^2 E_-)^*$ plus non-triviality (the contraction map of a non-degenerate alternating trilinear coupling, taken as data).
- *Stage D* (YukawaTrilinearForm) — introduced in v6, with operational predicates renamed in v7 — an alternating trilinear form $\omega : \Lambda^3 E_- \rightarrow K$ as primary datum, together with two explicit Prop predicates `ContractionInjective` and `ContractionSurjective`. From these the contraction map is *constructed* (via `alternatingMapLinearEquiv` and `AlternatingMap.curryLeft`) and proved to be a linear iso when both predicates hold and E_- is non-trivial, producing Stage C as a *theorem*, not as a definition.

Each upstream stage produces the next via a Lean bridge ($D \Rightarrow C \Rightarrow B \Rightarrow A$). Stage D carries the genuine content of TFPT Paper 2's "non-degenerate primitive alternating trilinear coupling": the trilinear form is the primary datum, the contraction iso is a derived fact, and the rank-3 conclusion follows from the contraction iso via the elementary dimension chain $\dim E_- = \dim(\Lambda^2 E_-)^* = \binom{\dim E_-}{2}$ on a non-trivial finite-dimensional space.

A seven-check `audit.sh` CI gate is included.

Not claimed here. No analytic content of Paper 1, no full geometric formalisation of $H^0(\mathbb{P}^1, \mathcal{O}(1))$, no quiver-theoretic definition of *primitive indecomposable Yukawa type*, no SM hypercharge spectrum on derived representations (duals, exterior powers, tensor products), no $\mathbb{Z}_6$ quotient, no α , no flavour, no cosmology.

Falsification or audit surface. The note fails if (i) `lake build` no longer succeeds against the pinned Mathlib release; (ii) the `audit.sh` script returns a non-zero exit; (iii) the dependency graph stated below drifts from the wording of Paper 2 §2.

Claim contract

Claim. The top-level Lean theorem `CarrierPremises.hypercharge_carrier_packet` takes four explicit interfaces:

- (1) a carrier involution element ε with $\varepsilon^2 = 1$ (`InvolutionProjectors.CarrierInvolution`, optionally wrapped by a `CalderonCertificate`);
- (2) a Higgs rank-2 certificate (`HiggsIndexCertificate`) discharging $\dim E_+ = 2$;
- (3) a Yukawa rank-3 witness (`YukawaRankCertificate`) discharging $\dim E_- = 3$, *refinable* via `YukawaTopForm` (Stage B), `PrimitiveIndecomposableYukawaCoupling` (Stage C), and `YukawaTrilinearForm` (Stage D) by formal Lean bridges;

- (4) a primitive oriented determinant-preserving carrier condition ($3q_- + 2q_+ = 0$, $q_- < 0 < q_+$, $\gcd(|q_-|, q_+) = 1$), packaged as `PrimitiveOrientedDeterminantCarrier 3 2`.

Conclusion. From these inputs Lean derives the $3 + 2$ hypercharge carrier $Y = -\frac{1}{3}P_- + \frac{1}{2}P_+$, with $(q_-, q_+) = (-2, 3)$, $\text{Tr } Y = 0$, and the carrier polynomial $6Y^2 - Y - 1 = 0$ in $\text{End}_K M$.

Three layers of the contract. The claim is structured into three explicit layers, separated to prevent overclaiming and to make every open question visible:

(a) *Direct inputs of CarrierPremises.* The bundled top-level theorem consumes the four interfaces above: a carrier involution, a Higgs rank witness, a Yukawa rank witness, and a primitive oriented determinant-preserving carrier condition.

(b) *Lean-proved upstream refinements (closed to the next algebraic layer).* Each of the four direct inputs is produced by an explicit Lean-proved upstream interface:

- Carrier involution $\leftarrow$ `CalderonProjector` $\leftarrow$ `BoundaryPolarization` ($H = H_+ \oplus H_-$). The converse `BoundaryPolarization.ofCarrierInvolution` makes the chain a Lean-proved *algebraic equivalence*: the carrier involution and the polarisation are two presentations of the same finite-dimensional carrier datum. This is not the analytic Paper-1 content; the analytic construction of the polarisation from boundary elliptic theory remains upstream.
- Higgs rank witness $\leftarrow$ `HiggsTopForm` (Stage B) $\leftarrow$ `HiggsSchemeCohomologyShadow` ($\simeq \text{MvPolynomial.homogeneousSubmodule (Fin 2) K 1}$).
- Yukawa rank witness $\leftarrow$ `YukawaTopForm` (Stage B) $\leftarrow$ `YukawaPrimitive` (Stage C) $\leftarrow$ `YukawaTrilinearForm` (Stage D) $\leftrightarrow$ `YukawaStageDExistence` (both directions formalised, under Nontrivial).
- Primitive oriented determinant carrier $\leftarrow$ `SeamWindingInterface` (algebraic packaging of seam-class data).

(c) *Open TFPT-origin theorems (not formalised here).* Three genuinely upstream questions remain open:

- (1) *Paper 1 analytic boundary theory* produces the polarisation $H = H_+ \oplus H_-$ (boundary elliptic theory, Sobolev spaces, Calderón operator).
- (2) *TFPT boundary kernel* canonically produces an $\omega : \Lambda^3 E_- \rightarrow K$ with `ContractionInjective` and `ContractionSurjective` (the *canonical* question; the existence is now closed by the present note).
- (3) *Seam winding* $[u_\Sigma] = 1$ produces a primitive oriented determinant-preserving carrier condition.

These three are *outside the scope* of the present Lean note; each is named, typed, and provided with a Lean-side target structure (`BoundaryPolarization`, `BoundaryYukawaKernelInterface`, `SeamWindingInterface`).

Not claimed. Beyond the three open TFPT-origin theorems, the $\mathbb{Z}_6$ quotient G_{phys} , derived SM representations (duals, exterior powers, tensor products), α , flavour, and cosmology are also not formalised in this note.

First assumptions on K . The projector construction requires 2 invertible in the carrier algebra; for vector-space applications this is ensured by $\text{char } K \neq 2$. The rational normalisation $Y = -P_-/3 + P_+/2$ further requires 6 invertible; concretely the bundled theorems use `CharZero K`.

Proof status. *Formally verified* in Lean 4 + Mathlib v4.29.1; no sorry; no admit; no unsafe/opaque/partial; no domain-specific axiom; `audit.sh` (seven checks including the signature-lock `AuditContract.lean`) returns AUDIT: PASS. As of v10, the three upstream interfaces (Calderón, Higgs, Yukawa Stage D backward) are each closed *to the next algebraic layer*. The remaining gaps are TFPT-side or analytic, not Lean-side.

Kill condition. Any of the seven audit checks fail; any headline theorem in `AuditCheck.lean` is removed or fails to elaborate; any example in `AuditContract.lean` fails to typecheck (which detects silent weakening of theorem statements).

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1 What This Note Is About

v1 formalised the *algebraic motor* of TFPT Paper 2 — the carrier polynomial and the trace identity on a concrete 5×5 diagonal model. v2 began making the upstream dependencies explicit. v3 made the dependency graph correct. v4 replaced the Yukawa rank witness with a structurally algebraic Stage B (top form on $\Lambda^3 E_-$). v5 introduced a Stage C contraction iso $E_- \simeq (\Lambda^2 E_-)^*$.

v6 completed the Yukawa refinement chain by introducing a Stage D primary datum; v7 renamed its predicates to operational names; v8 added a Calderón idempotent layer and a Higgs Stage B; v9 consolidates the document into a single review-ready state and adds an unconditional transparency theorem for Stage D. The Stage D primary datum reads:

$$\text{YukawaTrilinearForm } K E_- := \{ \omega : \Lambda^3 E_- \rightarrow_{\ell K} K \}, \quad \text{ContractionInjective } \omega, \text{ ContractionSurjective } \omega$$

The contraction map $C_\omega : E_- \rightarrow (\Lambda^2 E_-)^*$ is now *constructed* from ω via `exteriorPower.alternatingMapLi` composed with `AlternatingMap.curryLeft`. The two predicates state exactly that C_ω is injective (mathematical reading: non-degeneracy) and surjective (mathematical reading: primitivity / no spectator). Together, on a non-trivial finite-dimensional space, they yield a Stage C structure as a theorem; the rank-3 conclusion then follows mechanically:

$$\omega \text{ trilinear} + \text{CI} + \text{CS} + \text{Nontrivial } E_- \xrightarrow{\text{contraction is iso}} \text{Stage C} \xrightarrow{\binom{n}{2}=n, n \neq 0} \dim E_- = 3.$$

The non-triviality hypothesis is essential: for $E_- = 0$ both predicates are vacuously satisfied (the contraction $0 \rightarrow 0$ is trivially bijective) but $\dim E_- = 0$. The Lean theorem

`finrank_E_eq_zero_or_three` (new in v9) records the unconditional disjunction $\dim E_- = 0 \vee \dim E_- = 3$ for transparency; the rank-3 slogan above is the case under `Nontrivial E_-`.

The full Yukawa tower in Lean is now

$$\text{Stage D} \Rightarrow \text{Stage C} \Rightarrow \text{Stage B} \Rightarrow \text{Stage A} \Rightarrow \dim E_- = 3,$$

with every step a Lean theorem and Stages B and C carrying characterisation theorems establishing equivalence to $\dim E_- = 3$.

v6 made the Paper 1 input explicit. A new `CalderonInterface.lean` introduces a `CalderonCertificate` structure that names the Calderón-induced carrier involution as a Lean datum and feeds it into the `InvolutionProjectors` chain. v8 upgrades this interface from documentary to a structural algebraic step: a new `CalderonProjector.lean` module shows that an idempotent element π in a ring with 2 invertible yields $\varepsilon^2 = 1$ via the algebraic identity $\varepsilon := 2\pi - 1$, $(2\pi - 1)^2 = 4\pi^2 - 4\pi + 1 = 1$. v10 adds the next upstream layer: a new `BoundaryPolarization.lean` module shows that any direct-sum decomposition $H = H_+ \oplus H_-$ of a K -module produces an idempotent endomorphism on H (via `Submodule.IsCompl.projection`) and feeds the Calderón chain. The remaining Paper-1 gap is therefore precisely the analytic content: producing the boundary spectral decomposition itself.

v8 also introduces a Stage-B Higgs interface (`HiggsTopForm.lean`) parallel to the Yukawa tower: the rank-2 conclusion $\dim E_+ = 2$ is derivable from a structural hypothesis identifying E_+ with the algebraic shadow of $H^0(\mathbb{P}^1, \mathcal{O}(1))$, namely the space of homogeneous degree-1 polynomials in two formal variables $\alpha X + \beta Y$. v10 closes the bridge to actual algebraic geometry: `HiggsSchemeCohomologyShadow.lean` provides a `LinearEquiv` between this algebraic shadow and the Mathlib-native degree-1 piece of the bivariate polynomial ring `MvPolynomial.homogeneousSubmodule (Fin 2) K 1 = K[X0, X1]1`, which by the standard Proj construction is the algebraic-geometry presentation of $H^0(\mathbb{P}_K^1, \mathcal{O}(1))$. The remaining gap is now the scheme-theoretic identification $H^0(\text{Proj } K[X, Y], \mathcal{O}(1)) \simeq K[X, Y]_1$, a general algebraic-geometry theorem independent of TFPT.

v10 also closes the Stage D backward direction (`YukawaStageDExistence.lean`). Given any basis $b : \text{Basis (Fin 3)} K E$, the determinant form $b.\text{det}$ transports via `exteriorPower.alternatingMapLinearE` to an alternating trilinear form $\omega : \Lambda^3 E \rightarrow_{\ell K} K$. The contraction C_ω is shown injective by basis-level matrix-determinant computation and surjective by $\dim E = \dim(\Lambda^2 E)^* = 3$ and the standard finite-dimensional injective-iff-surjective theorem. From this the Stage D characterisation $\exists \omega (CI \wedge CS) \iff \dim E = 3$ now holds (under `Nontrivial E`).

The bundled main theorem packages the three load-bearing conclusions

$$(q_-, q_+) = (-2, 3), \quad \text{Tr } Y = 0, \quad 6Y^2 - Y - 1 = 0$$

into a single Lean statement `hypercharge_carrier_packet` that consumes only structured hypotheses — with the Yukawa input now refinable all the way to Stage D, and the boundary input now arriving via an explicit Calderón certificate.

The redesign is governed by one editorial discipline: *the language must not outrun the formal status*. Two examples:

- This note never claims to formalise the *Standard- Model hypercharge spectrum*, only the $3 + 2$ carrier multiplet. Duals, exterior powers, tensor products — the constructions that lift the carrier to all SM chiral multiplets — are out of scope (§7).
- The Yukawa Stage-A certificate is named `YukawaRankCertificate`, not `PrimitiveYukawaTypeCertificate`. The certificate is a *rank witness*; it is not a formalisation of "primitive indecomposable Yukawa type", which is the proper open problem.

2 What Was Built

Twenty-seven Lean modules under `experiments/lean4-carrier-rigidity/TfptCarrier/`:

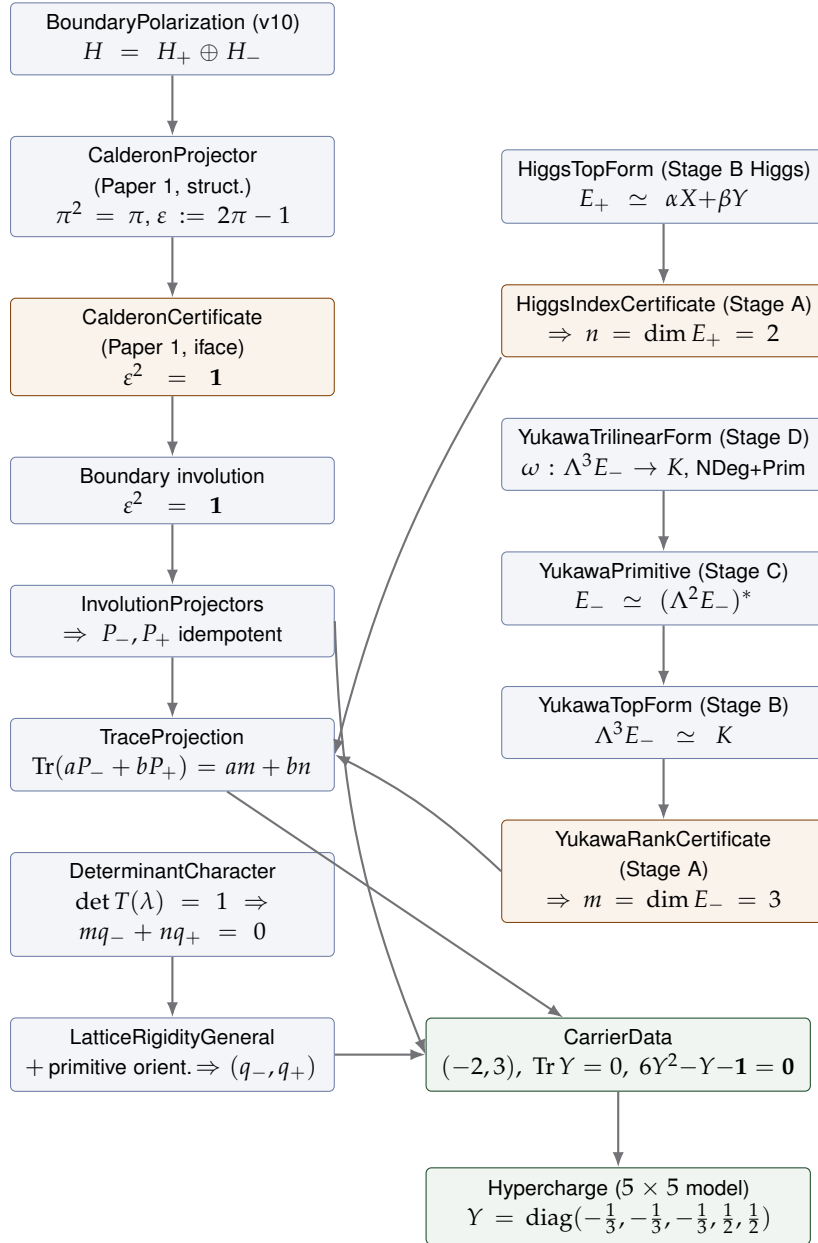
Module	Lines	Content
<code>Polarization.lean</code>	~ 150	Orthogonal-idempotent algebra; $\text{six}Y^2 - \text{six}Y - 6 = 0$.
<code>InvolutionProjectors.lean</code>	~ 200	$\varepsilon^2 = 1 \Rightarrow P_-, P_+$ complete orthogonal idempotents.
<code>CalderonInterface.lean</code>	~ 110	Paper 1 <code>CalderonCertificate</code> : explicit interface wrapper feeding <code>CarrierInvolution</code> .
<code>CalderonProjector.lean</code>	~ 125	Structural derivation: idempotent $\pi^2 = \pi \Rightarrow \varepsilon := 2\pi - 1$ satisfies $\varepsilon^2 = 1$.
<code>BoundaryPolarization.lean</code>	~ 200	v10 upstream: $H = H_+ \oplus H_-$ produces an idempotent endomorphism on H via <code>Mathlib's IsCompl.projection</code> ; feeds <code>CalderonProjector</code> .
<code>MathlibBridge.lean</code>	~ 55	<code>Polarization</code> $\leftrightarrow$ <code>CompleteOrthogonalIdempotents</code> .
<code>LatticeRigidityGeneral.lean</code>	~ 200	General (m, n) primitive rigidity; SM corollary $(3, 2) \rightarrow (-2, 3)$.
<code>Rigidity.lean</code>	~ 40	SM-case wrapper.
<code>OrientedDeterminantCarrier.lean</code>	~ 100	<code>PrimitiveOrientedDeterminantCarrier (m n)</code> : trace-zero + orientation + primitivity bundled.
<code>TraceProjection.lean</code>	~ 100	$\text{Tr}(aP_- + bP_+) = a \dim E_- + b \dim E_+$; trace-zero corollary.
<code>DeterminantCharacter.lean</code>	~ 130	$\det T(\lambda) = \lambda^{mq_- + nq_+}$; universal- λ and at-two headlines.
<code>HiggsIndexShadow.lean</code>	~ 125	Stage-A <code>HiggsIndexCertificate</code> ; algebraic shadow has dim 2.
<code>HiggsTopForm.lean</code>	~ 210	Stage-B <code>HiggsTopForm</code> : $E_+ \simeq$ space of $\alpha X + \beta Y$ (degree-1 in two formal vars), $\dim = 2$ derived.
<code>HiggsSchemeCohomologyShadow.lean</code>	~ 200	v10 upstream: $\text{DegreeOneTwoVarForm } K \simeq_{\ell K} \text{MvPolynomial.homogeneousSubmodule (Fin 2) } K 1 (= K[X_0, X_1]_1, \text{Proj-side } H^0)$.
<code>YukawaRank.lean</code>	~ 135	Stage-A <code>YukawaRankCertificate</code> (rank witness, $E_- \simeq K^3$).
<code>YukawaTopForm.lean</code>	~ 195	Stage-B <code>YukawaTopForm</code> ($\Lambda^3 E_- \simeq K$) + rank-3 theorem + characterisation.
<code>YukawaPrimitive.lean</code>	~ 215	Stage-C <code>PrimitiveIndecomposableYukawaCoupling</code> ($E_- \simeq (\Lambda^2 E_-)^*$) + rank-3 theorem + characterisation.
<code>YukawaTrilinearForm.lean</code>	~ 210	Stage-D <code>YukawaTrilinearForm</code> ($\omega : \Lambda^3 E_- \rightarrow K$) + operational <code>ContractionInjective/ContractionSurjective</code> predicates + constructed contraction + rank-3 theorem (under <code>Nontrivial E</code>) + unconditional $0 \vee 3$ transparency theorem.
<code>YukawaStageDExistence.lean</code>	~ 280	v10 Stage D backward: $\dim E = 3 \Rightarrow \exists \omega$ with <code>CI</code> $\wedge$ <code>CS</code> via <code>b.det</code> + direct 3×3 matrix-determinant computation.
<code>BoundaryYukawaKernelInterface.lean</code>	~ 200	v11 abstract typed target: <code>BoundaryYukawaKernel</code> packages $(\omega, \text{CI}, \text{CS}, \text{Nontrivial})$ with characterisation $\iff \dim E_- = 3$.
<code>SeamWindingInterface.lean</code>	~ 130	v11 abstract typed target: <code>SeamWindingData (m, n)</code> packages the seam-class consequences (trace-zero, orientation, primitivity); bidirectional bridge to <code>PrimitiveOrientedDeterminantCarrier</code> .
<code>CarrierData.lean</code>	~ 200	Bundled main theorems: pair, trace, polynomial, packet.
<code>Hypercharge.lean</code>	~ 95	Concrete 5×5 model: $\text{Tr } Y = 0$ and $6Y^2 - Y - 1 = 0$.
<code>Sanity.lean</code>	30	<code>#eval</code> smoke tests.
<code>AxiomCheck.lean</code>	80	<code>#print</code> axioms on every headline theorem.
<code>AuditCheck.lean</code>	80	<code>#check</code> elaboration of every headline name.
<code>AuditContract.lean</code>	~ 180	Signature-lock example patterns for every headline theorem (strict signature audit).

Plus:

- `scripts/audit.sh` — seven-check hard CI gate (`build` / `sorry` / `axiom` / `unsafe-opaque-partial` / `axiom-print` / `headline-#check` / `signature-lock example`).
- `lakefile.lean` pinning Lean 4.29.1 and Mathlib v4.29.1.

3 Dependency Graph

Every node is a separately formalised Lean theorem; every arrow is a discharged proof obligation in TtptCarrier. Blue nodes are unconditional Lean theorems. Orange nodes are Stage-A structured hypotheses. Green nodes are the bundled conclusions.



Three things to note about the graph that the previous version got wrong, and one new feature:

- **No** `InvolutionProjectors` $\rightarrow$ `LatticeRigidityGeneral` **arrow**. The projector construction does not imply lattice rigidity; the two branches are independent until they meet at `CarrierData`.
- **The arrow runs** `DeterminantCharacter` $\rightarrow$ `LatticeRigidityGeneral`, **not the reverse**. The determinant character produces the trace-zero equation $mq_- + nq_+ = 0$; the rigidity theorem then upgrades it to a unique pair when combined with the primitivity and sign conditions.

- **No** `HiggsIndexCertificate` $\rightarrow$ `YukawaRankCertificate` **arrow**. The two certificates discharge different ranks via independent mathematical arguments. Both feed into `CarrierData` separately.
- **New in v4:** `YukawaTopForm` $\rightarrow$ `YukawaRankCertificate` **arrow**. The Yukawa input was split into two layers. Stage B (`YukawaTopForm`) is the algebraic statement $\Lambda^3 E_- \simeq K$, which is the precise Lean reading of TFPT Paper 2's " $\Lambda^3 E_- = \det E_-$ ". The Stage A rank witness drops out as a corollary via an arbitrary basis.
- **New in v5:** `YukawaPrimitive` $\rightarrow$ `YukawaTopForm` **arrow**. The Yukawa input is further refined into a Stage C `PrimitiveIndecomposableYukawaCoupling`: a contraction iso $E_- \simeq (\Lambda^2 E_-)^*$ plus non-triviality.
- **New in v6:** `YukawaTrilinearForm (Stage D)` $\rightarrow$ `YukawaPrimitive` **arrow**. Stage D introduces an alternating trilinear form $\omega : \Lambda^3 E_- \rightarrow K$ as primary datum, together with two explicit operational Prop predicates (`ContractionInjective` and `ContractionSurjective` from v7 onwards, originally `Nondegenerate` and `Primitive` in v6). The contraction C_ω is *constructed* from ω (no longer data), and the iso conclusion of Stage C becomes a Lean theorem under the two predicates and `Nontrivial E_-`. This is the upstream-most layer of the Yukawa tower; the rank-3 conclusion is now derived from an alternating trilinear form together with two structural predicates and non-triviality.
- **New in v6:** `CalderonCertificate` $\rightarrow$ `CarrierInvolution` **arrow**. The Paper 1 input is no longer silently consumed: a `CalderonCertificate` is a thin wrapper that names the Calderón-induced carrier involution as a Lean datum, with a forgetful map into the `InvolutionProjectors` chain.
- **New in v7:** `AuditContract.lean` **signature locks**. Each headline theorem is used in an example : `<exact intended type> := <theorem>` pattern, detecting silent weakening of theorem statements that `#check` would miss.
- **New in v7:** `PrimitiveOrientedDeterminantCarrier (m n)`. The three-part TFPT carrier condition (trace-zero, orientation, primitivity) is bundled as a single Lean structure, removing the semantic-drift risk of an unstructured triple of hypotheses.
- **New in v8:** `CalderonProjector` $\rightarrow$ `CalderonCertificate` **arrow**. The algebraic step $\pi^2 = \pi \Rightarrow \varepsilon^2 = 1$ (for $\varepsilon := 2\pi - 1$) is now a Lean theorem, upgrading the Calderón layer from documentary interface to a structural algebraic derivation.
- **New in v8:** `HiggsTopForm` $\rightarrow$ `HiggsIndexCertificate` **arrow**. A Higgs Stage B structure identifies E_+ with the algebraic shadow of $H^0(\mathbb{P}^1, \mathcal{O}(1))$, namely `DegreeOneTwoVarForm K`, and derives $\dim E_+ = 2$ from this polynomial-space iso rather than from a bare rank witness.
- **New in v9:** `finrank_E_eq_zero_or_three`. Unconditional Stage D transparency theorem: without `Nontrivial E` the operational predicates force only $\dim E = 0 \vee \dim E = 3$, which is the precise mathematical reason the rank-3 slogan must carry the `Nontrivial E` hypothesis at the use site.
- **New in v10:** `BoundaryPolarization` $\rightarrow$ `CalderonProjector` **arrow**. Any $H = H_+ \oplus H_-$ gives an idempotent endomorphism on H via Mathlib's `IsCompl.projection`. The Calderón input is now derived from a polarisation datum (one step closer to Paper 1's analytic source).
- **New in v10:** `HiggsSchemeCohomologyShadow` $\rightarrow$ `HiggsTopForm` **arrow**. The Higgs Stage-B shadow is pinned to Mathlib's `MvPolynomial.homogeneousSubmodule (Fin 2) K 1`, i.e.

the Proj-side degree-1 piece of $K[X_0, X_1]$.

- **New in v10:** `YukawaStageDExistence`. Stage D backward direction: $\dim E = 3 \Rightarrow \exists \omega \text{ (CI} \wedge \text{CS)}$. The Stage D characterisation $\exists \omega \iff \dim E = 3$ now holds under `Nontrivial E`.

4 The Headline Theorems

We list the headline theorems by graph layer. Every name is elaborated by `TfptCarrier/AuditCheck.lean`.

Layer 1 — Polarization (algebraic core)

Theorem 4.1 (`Polarization.sixY_carrier_polynomial`). *Let A be a ring with two orthogonal idempotents P_-, P_+ summing to $\mathbf{1}$. Then $\text{sixY} := -2P_- + 3P_+$ satisfies $\text{sixY}^2 - \text{sixY} - 6 = \mathbf{0}$.*

Layer 0+ — Calderón structural certificate (Paper 1)

Theorem 4.2 (`CalderonProjector.eps_sq`). *Let A be a ring with 2 invertible and let $\pi \in A$ satisfy $\pi^2 = \pi$. Define $\varepsilon := 2\pi - 1$. Then $\varepsilon^2 = \mathbf{1}$.*

Theorem 4.3 (`CalderonProjector.toCalderonCertificate`). *Every `CalderonProjector` produces a `CalderonCertificate` (the Paper 1 interface). The Calderón entry point is now derived from a single algebraic hypothesis — the existence of the boundary-spectral idempotent.*

Layer 2 — Involution to projectors

Theorem 4.4 (`InvolutionProjectors.CarrierInvolution.toPolarization`). *Let ε be an element of a ring A with $\varepsilon^2 = \mathbf{1}$ and 2 invertible in A . The spectral projectors P_-, P_+ defined by $P_{\pm} := (\mathbf{1} \pm \varepsilon) \cdot \hat{2}$ form a *Polarization* of A . Each of the four polarization axioms is discharged directly from $\varepsilon^2 = \mathbf{1}$.*

Layer 3 — General lattice rigidity

Theorem 4.5 (`LatticeRigidityGeneral.primitive_trace_free_pair_general`). *Let m, n be positive integers and $q_-, q_+ \in \mathbb{Z}$ satisfy*

$$mq_- + nq_+ = 0, \quad q_- < 0 < q_+, \quad \gcd(|q_-|, q_+) = 1.$$

Then $q_- = -n / \gcd(m, n)$ and $q_+ = m / \gcd(m, n)$.

Corollary 4.6 (`Rigidity.unique_carrier_pair`). *For $(m, n) = (3, 2)$ the same three hypotheses force $(q_-, q_+) = (-2, 3)$.*

Layer 4 — Structural trace

Theorem 4.7 (`TraceProjection.trace_linear_combination_of_idempotents`). *For any two idempotent endomorphisms P_-, P_+ of a finite-dimensional K -vector space M , and any $a, b \in K$,*

$$\text{Tr}_M(a \cdot P_- + b \cdot P_+) = a \cdot \dim_K(\text{range } P_-) + b \cdot \dim_K(\text{range } P_+).$$

Corollary 4.8 (`TraceProjection.trace_carrier_Y_eq_zero`). *With $a = -\frac{1}{3}, b = \frac{1}{2}$ and the rank discharges $\dim E_- = 3, \dim E_+ = 2$, the structural identity gives $\text{Tr } Y = 0$ for $Y = -\frac{1}{3}P_- + \frac{1}{2}P_+$. The hypothesis `[CharZero K]` is genuinely used here.*

Layer 5 — Determinant character

Theorem 4.9 (`DeterminantCharacter.det_torusMatrix`). *For nonzero λ in a field K and integer charges q_-, q_+ ,*

$$\det T(\lambda) = \lambda^{m \cdot q_- + n \cdot q_+}, \quad T(\lambda) := \text{diag}(\underbrace{\lambda^{q_-}, \dots, \lambda^{q_-}}_m, \underbrace{\lambda^{q_+}, \dots, \lambda^{q_+}}_n).$$

Corollary 4.10 (`DeterminantCharacter.trace_zero_of_det_one_forall_rat`). If $\det T(\lambda) = 1$ for every positive rational $\lambda \neq 1$, then $m q_- + n q_+ = 0$. The single-point form $\det T(2) = 1 \Rightarrow m q_- + n q_+ = 0$ is recorded separately as `trace_zero_of_det_one_at_two`.

Remark. The Diophantine equation $m q_- + n q_+ = 0$ from the determinant character is necessary but *not sufficient* to fix (q_-, q_+) : in $(m, n) = (3, 2)$ it is satisfied by every pair of the form $(-2k, 3k)$. The primitivity ($\gcd = 1$) and sign ($q_- < 0 < q_+$) conditions are what reduce that infinite family to the unique pair $(-2, 3)$. The combined hypothesis is therefore phrased throughout as the *primitive oriented determinant-preserving carrier condition*.

Layer 6 — Stage-A and Stage-B rank certificates

Theorem 4.11 (`HiggsTopForm.DegreeOneTwoVarForm.finrank_eq_two`). The space of homogeneous degree-1 polynomials in two formal variables, defined concretely as `DegreeOneTwoVarForm K := {(\alpha, \beta) : K \times K}` with the natural K -module structure, has K -rank 2.

Theorem 4.12 (`HiggsTopForm.HiggsTopForm.finrank_Eplus_eq_two`). A Higgs Stage-B certificate (a linear equivalence $E_+ \simeq \text{DegreeOneTwoVarForm } K$) forces $\dim_K E_+ = 2$. The Stage A $\rightarrow$ Stage B refinement is a real algebraic statement: E_+ is identified with the polynomial space, not just with K^2 .

Layer 6 — Stage-A rank certificates

Theorem 4.13 (`HiggsIndexShadow.HiggsIndexCertificate.finrank_Eplus_eq_two`). A *HiggsIndexCertificate* defined as a linear equivalence $E_+ \simeq H^0(\mathbb{P}^1, \mathcal{O}(1))^{\text{alg}} \simeq K^2$, forces $\dim_K E_+ = 2$.

Theorem 4.14 (`YukawaRank.YukawaRankCertificate.finrank_Eminus_eq_three`). A *YukawaRankCertificate* (Stage A), defined as a linear equivalence $E_- \simeq K^3$, forces $\dim_K E_- = 3$.

Layer 6b — Yukawa Stage B (algebraic content)

Theorem 4.15 (`YukawaTopForm.finrank_lambda3_eq_one`). A *YukawaTopForm* $K \rightarrow E$, defined as a linear equivalence $\Lambda^3 E \simeq K$, forces $\dim_K (\Lambda^3 E) = 1$.

Theorem 4.16 (`YukawaTopForm.finrank_E_eq_three`). For a finite-dimensional K -vector space, a *YukawaTopForm* forces $\dim_K E = 3$. The proof is the composition

$$\Lambda^3 E \simeq K \Rightarrow \dim \Lambda^3 E = 1 \xrightarrow{\text{finrank_eq}} \binom{\dim E}{3} = 1 \xrightarrow{\text{choose_eq_one_iff}} \dim E = 3.$$

Theorem 4.17 (`YukawaTopForm.ofFinrankEqThree`). Conversely, any finite-dimensional K -vector space with $\dim E = 3$ admits a *YukawaTopForm*: pick any basis and read off the resulting one-dimensional $\Lambda^3 E$ as K .

Theorem 4.18 (`YukawaTopForm.nonempty_iff_finrank_eq_three`). Characterisation: for a finite-dimensional K -vector space,

$$\text{Nonempty } (\text{YukawaTopForm } K \rightarrow E) \iff \dim_K E = 3.$$

The Stage B hypothesis is therefore precisely the rank-3 condition, expressed in exterior-algebra form.

Theorem 4.19 (`YukawaTopForm.toYukawaRankCertificate`). A Stage B *YukawaTopForm* produces a Stage A *YukawaRankCertificate* by extracting $\dim E = 3$ and choosing an arbitrary basis.

Layer 6c — Yukawa Stage C (primitive indecomposable coupling)

Theorem 4.20 (`PrimitiveIndecomposableYukawaCoupling.finrank_E_eq_three`). A *PrimitiveIndecomposable* K E , defined as a linear equivalence $E \simeq (\Lambda^2 E)^*$ plus non-triviality of E , forces $\dim_K E = 3$.

Proof sketch. The contraction iso transports dimensions as

$$\dim E = \dim(\Lambda^2 E)^* \stackrel{\text{dual}}{=} \dim \Lambda^2 E \stackrel{\text{exteriorPower}}{=} \binom{\dim E}{2} = \frac{n(n-1)}{2}.$$

The integer equation $n = n(n-1)/2$ with $n \geq 1$ has the unique solution $n = 3$ (a finite case split / omega).

Theorem 4.21 (`PrimitiveIndecomposableYukawaCoupling.toYukawaTopForm`). A Stage C coupling produces a Stage B *YukawaTopForm* via the v4 converse *YukawaTopForm.ofFinrankEqThree* applied to *finrank_E_eq_three*.

Theorem 4.22 (`PrimitiveIndecomposableYukawaCoupling.toYukawaRankCertificate`). A Stage C coupling produces a Stage A *YukawaRankCertificate* through the composition $C \Rightarrow B \Rightarrow A$.

Theorem 4.23 (`PrimitiveIndecomposableYukawaCoupling.ofFinrankEqThree`). Conversely, any finite-dimensional K -vector space with $\dim E = 3$ admits a Stage C coupling: a contraction iso $E \simeq (\Lambda^2 E)^*$ exists by dimension count, since both sides have dimension 3 (via *dual_finrank_eq* and *exteriorPower.finrank_eq*).

Theorem 4.24 (`PrimitiveIndecomposableYukawaCoupling.nonempty_iff_finrank_eq_three`). For a finite-dimensional K -vector space, $\text{Nonempty}(\text{PrimitiveIndecomposableYukawaCoupling } K E) \iff \dim_K E = 3$.

Layer 6d — Yukawa Stage D (primary trilinear form)

Theorem 4.25 (`YukawaTrilinearForm.contraction`). For every alternating trilinear form $\omega : \Lambda^3 E \rightarrow K$, the associated contraction map

$$C_\omega : E \rightarrow (\Lambda^2 E)^*, \quad C_\omega(v)(w \wedge x) := \omega(v \wedge w \wedge x),$$

is constructible as a *LinearMap*. The Lean construction uses *exteriorPower.alternatingMapLinearEquiv* (to convert between $(\Lambda^n E)^*$ and the space of K -valued alternating maps on E^n) and *AlternatingMap.curryLeft* (to split off the first argument).

Theorem 4.26 (`YukawaTrilinearForm.ContractionInjective` and `ContractionSurjective`). Two explicit operational *Prop* predicates on ω :

$$\text{ContractionInjective } \omega := \text{Injective } C_\omega, \quad \text{ContractionSurjective } \omega := \text{Surjective } C_\omega.$$

The TFPT-side phrases “non-degenerate” and “primitive / no spectator” are docstring synonyms for the two operational predicates respectively. The combined predicate *PrimitiveIndecomposableYukawaCondition* bundles both (plus *Nontrivial E* at the use site).

Theorem 4.27 (`YukawaTrilinearForm.contractionEquivOfBoth`). A trilinear form ω satisfying both predicates over a finite-dimensional vector space induces a linear iso $E \simeq (\Lambda^2 E)^*$ — the contraction C_ω itself, upgraded to a *LinearEquiv* via *LinearEquiv.ofBijective*.

Theorem 4.28 (`YukawaTrilinearForm.toPrimitiveIndecomposableYukawaCoupling`). A trilinear form with *ContractionInjective* and *ContractionSurjective* on a non-trivial finite-dimensional vector space yields a Stage C *PrimitiveIndecomposableYukawaCoupling*. Together with the Stage $C \Rightarrow \text{Stage B} \Rightarrow \text{Stage A}$ chain, this means

$$\omega : \Lambda^3 E \rightarrow K \text{ (alt.) plus } CI, CS, \text{Nontrivial } E \implies \dim E = 3.$$

Theorem 4.29 (`YukawaTrilinearForm.finrank_E_eq_zero_or_three (v9)`). Unconditional transparency form, no `Nontrivial` hypothesis. For any finite-dimensional K -vector space E , a trilinear form ω satisfying `ContractionInjective` and `ContractionSurjective` forces

$$\dim_K E = 0 \quad \vee \quad \dim_K E = 3.$$

The two predicates are vacuously satisfied when $E = 0$; the rank-3 conclusion holds precisely when E is non-trivial. This theorem records the fact that the operational predicates do not themselves rule out the trivial case, which is the mathematical reason the rank-3 slogan above carries the `Nontrivial E` hypothesis.

What Stage D says about TFPT Paper 2. The phrase "primitive indecomposable Yukawa type" in TFPT Paper 2 contains three semi-formal qualifiers:

- *primitive* — the trilinear coupling does not factor through a sub-trilinear form;
- *indecomposable* — the carrier does not split as a direct sum that respects the coupling;
- *no spectator* — the closure $E_- \otimes \Lambda^2 E_- \rightarrow \Lambda^3 E_-$ exhausts E_- .

Lean translation in Stage D. The primary datum is the alternating trilinear form ω on E_- . The three qualifiers translate to two explicit operational Prop predicates on ω : `ContractionInjective` (the contraction is injective; equivalently, no nonzero v has $C_\omega(v) = 0$ — *non-degenerate* in TFPT-side prose) and `ContractionSurjective` (the contraction is surjective; equivalently, every linear functional on $\Lambda^2 E_-$ is the contraction of ω with some vector — *primitive / no spectator* in TFPT-side prose). Together with `Nontrivial E_-` they imply the Stage C contraction iso and hence $\dim E_- = 3$.

Where the standing Yukawa gap now lives. The Lean chain takes the trilinear form ω together with the two predicates and `Nontrivial E_-` as input. The remaining open question is *purely upstream*: does the TFPT boundary kernel (`CalderonCertificate` plus representation-theoretic data that the Yukawa coupling exists) produce a trilinear form satisfying `ContractionInjective` and `ContractionSurjective`? That question is TFPT theory work, not Lean engineering. The Lean chain is now flat: there is no hidden assumption between Stage D and $\dim E_- = 3$.

Layer 7 — Mathlib bridge

Theorem 4.30 (`MathlibBridge.Polarization.toCompleteOrthogonalIdempotents`). A *TFPT.Carrier.Polarization* on a ring A is canonically a *Mathlib CompleteOrthogonalIdempotents* on the pair (P_-, P_+) . A converse map of `CompleteOrthogonalIdempotents` is also provided.

Layer 8 — Bundled main theorems

The carrier-rigidity payload of the project is the following set of bundled theorems on `CarrierPremises K M`:

Theorem 4.31 (`CarrierPremises.primitive_pair_eq_sm`). Under the primitive oriented determinant-preserving carrier condition with rank data $(m, n) = (3, 2)$, $(q_-, q_+) = (-2, 3)$.

Theorem 4.32 (`CarrierPremises.trace_Y_eq_zero`). $\text{Tr}_M Y = 0$ for $Y = -\frac{1}{3}P_- + \frac{1}{2}P_+$.

Theorem 4.33 (`CarrierPremises.carrier_polynomial_Y`). $6 \cdot (Y \cdot Y) - Y - \mathbf{1} = \mathbf{0}$ in $\text{End}_K M$. The hypothesis $[\text{CharZero } K]$ is used here to provide $\widehat{6} \in K$.

Theorem 4.34 (`CarrierPremises.hypercharge_carrier_packet`). Under *CarrierPremises* and the primitive oriented determinant-preserving carrier condition, the three bundled conclusions above hold simultaneously. This is the top-level claim of the project.

Concrete realisation

Theorem 4.35 (Hypercharge.trace_Y, Hypercharge.Y_carrier_polynomial). *On the concrete 5×5 rational diagonal model, $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ satisfies $\text{Tr } Y = 0$ and $6Y^2 - Y - \mathbf{1} = \mathbf{0}$, proved by direct per-entry rational arithmetic.*

5 Significance

The two load-bearing chains, with all hypotheses explicitly visible, are:

Chain A — trace identity.

Calderón certificate $\xrightarrow{\text{Paper 1 interface}} \varepsilon^2 = \mathbf{1} \xrightarrow{\text{InvolutionProjectors}} P_-, P_+$ orthogonal idempotents

together with the rank discharges

$\dim E_+ = 2$ (HiggsIndexCertificate), $\omega : \Lambda^3 E_- \rightarrow K + \text{CI} + \text{CS} + \text{Nontrivial}$ (Stage D) $\Rightarrow \dim E_- = 3$

and the determinant-normalised coefficient choice $Y := -\frac{1}{3}P_- + \frac{1}{2}P_+$ give, via TraceProjection,

$$\text{Tr } Y = -\frac{1}{3} \cdot 3 + \frac{1}{2} \cdot 2 = 0.$$

The Yukawa input is now an alternating trilinear form with two explicit Prop predicates. The contraction iso and the rank-3 conclusion are both derived theorems.

Chain B — integer rigidity. The trace-zero condition

$$m q_- + n q_+ = 0$$

follows from the universal- λ determinant character. Combined with the primitivity and orientation conditions

$$q_- < 0 < q_+, \quad \gcd(|q_-|, q_+) = 1,$$

the general (m, n) rigidity theorem gives

$$q_- = -\frac{n}{\gcd(m, n)}, \quad q_+ = \frac{m}{\gcd(m, n)}.$$

For $(m, n) = (3, 2)$ this specialises to $(q_-, q_+) = (-2, 3)$. The integer answer is a corollary of a parametric theorem; it is not hard-coded.

Why both chains matter together. The carrier polynomial $6Y^2 - Y - \mathbf{1} = \mathbf{0}$ is a consequence of the polarization layer alone (it does not need either rank discharge). The trace identity $\text{Tr } Y = 0$ requires the rank discharges. The integer pair $(-2, 3)$ requires the primitive oriented determinant-preserving condition together with $(m, n) = (3, 2)$. The bundled theorem `hypercharge_carrier_packet` is the unique statement that requires all upstream pieces simultaneously.

What is *not* claimed. The $3 + 2$ multiplet $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ is the carrier. The *full Standard-Model hypercharge spectrum* — all chiral-multiplet hypercharges including duals, exterior powers, and tensor products — is not formalised in this experiment. The title and main theorem both say "3 + 2 hypercharge carrier", not "SM hypercharge spectrum".

6 Interface Table: TFPT Paper 2 $\leftrightarrow$ Lean Status

The following table maps each load-bearing claim of TFPT Paper 2 to its current Lean status and the still-missing upstream proof. It is the primary defence against scope-creep: the present note formalises columns 1 and 2; column 3 is explicitly out of scope.

TFPT Paper 2 claim	Lean status (v11)	Missing upstream proof
$\varepsilon^2 = 1$ on E	Two structural layers: <code>CalderonProjector</code> (idempotent π gives $\varepsilon := 2\pi - 1$ with $\varepsilon^2 = 1$) and <code>BoundaryPolarization</code> (any $H = H_+ \oplus H_-$ gives such an idempotent via <code>IsCompl.projection</code>); feeds <code>CalderonCertificate</code> .	Analytic construction of a boundary spectral decomposition $H = H_+ \oplus H_-$ from Paper 1's elliptic Calderón operator on Sobolev spaces.
$\dim E_+ = 2$	Stages A, B, and B+ all formalised (<code>HiggsIndexCertificate</code> , <code>HiggsTopForm</code> , <code>HiggsSchemeCohomologyShadow</code>); <code>DegreeOneTwoVarForm K</code> $\simeq_{\ell K}$ <code>MvPolynomial.homogeneousSubmodule</code> (Fin 2) $K \mathbf{1} = K[X_0, X_1]_1$, the algebraic-geometry "Proj-side" presentation of $H^0(\mathbb{P}^1, \mathcal{O}(1))$	Scheme-theoretic identification $H^0(\text{Proj } K[X, Y], \mathcal{O}(1)) \simeq K[X, Y]_1$ (a general algebraic-geometry theorem, independent of TFPT).
$\dim E_- = 3$	Stage A–D forward + Stage D backward (<code>YukawaStageDExistence</code> , v10): $\dim E_- = 3 \Rightarrow \exists \omega$ with <code>ContractionInjective</code> and <code>ContractionSurjective</code> , via <code>b.det</code> and direct 3×3 matrix computation. The Stage D characterisation $\exists \omega (CI \wedge CS) \iff \dim E_- = 3$ holds under <code>Nontrivial E_-</code> .	TFPT-internal construction of ω from boundary kernel and representation data (the existence is now algebraically closed by basis-determinant; what remains is whether TFPT's boundary derivation produces <i>this particular</i> ω).
$3q_- + 2q_+ = 0$	Theorem from determinant character of carrier torus (<code>DeterminantCharacter</code>)	Seam-winding $[u_\Sigma] = 1$ produces the determinant-preserving condition.
$(q_-, q_+) = (-2, 3)$	Theorem from general (m, n) lattice rigidity (<code>LatticeRigidityGeneral</code>) with <code>PrimitiveOrientedDeterminantCarrier</code> as input	Orientation $q_- < 0 < q_+$ and primitivity $\gcd = 1$ from seam-class data.
$Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$	Bundled theorem (<code>CarrierPremises.hypercharge_carrier_packet</code>) plus concrete 5×5 model (<code>Hypercharge</code>)	None (this is the conclusion).
$\text{Tr}_E Y = 0$	Bundled theorem (<code>CarrierPremises.trace_Y_eq_zero</code>)	None.
$6Y^2 - Y - \mathbf{1} = \mathbf{0}$	Bundled theorem (<code>CarrierPremises.carrier_polynomial_Y</code>) in $\text{End}_K M$	None.
$S^+ = \Lambda^{\text{even}} E$, family ν^c	<i>Not formalised</i>	Exterior-power packet construction; chiral-family decomposition.
$N_{\text{fam}} = 3$	<i>Not formalised</i>	Seam winding $\rightarrow$ family count.
$G_{\text{phys}} = (SU(3) \times SU(2) \times U(1)_Y) / \mathbb{Z}_6$	<i>Not formalised</i>	Stabilizer theorem for the carrier datum; $\mathbb{Z}_6$ quotient theorem.
SM hypercharges on all chiral multiplets	<i>Not formalised</i>	Derived representations: dual, exterior powers, tensor products of the carrier.
$b_1 = 41/10$, $\Omega_{\text{adm}} = 48$, $N_\Phi = 1$	<i>Not formalised</i>	Index ledger; representation audit.

The pattern is clear. The Lean note formalises the *carrier-rigidity algebra* of Paper 2 in full;

the *TFPT-internal derivations* feeding that algebra remain upstream work; and the *downstream SM-packet content* (derived representations, $\mathbb{Z}_6$ quotient, family counting, index ledger) is intentionally out of scope.

7 Limitations

Yukawa: Stage D resolves the v5 critique. The v5 critique of Stage C was that it bundled the contraction iso as data, so it was equivalent (in finite dimensions) to $\dim E = 3$ and did not "explain" the rank. v6 resolved this by introducing Stage D; v7 renamed its operational predicates to keep semantic-load words out of the Lean interface:

$$\underbrace{\omega : \Lambda^3 E_- \rightarrow K}_{\text{primary datum}} \underbrace{+ \text{ContractionInjective} + \text{ContractionSurjective} + \text{Nontrivial } E_-}_{\text{operational predicates and non-triviality}} \xrightarrow{\text{Lean theorem}} \dim$$

The contraction C_ω is a *constructed* linear map; the iso conclusion is a Lean theorem under the two predicates and non-triviality of E_- ; the rank-3 statement follows from the iso via elementary dimension counting. There is no longer any structural ambiguity at the Yukawa interface in Lean.

Why Nontrivial E_- is necessary. Without the non-triviality hypothesis the rank-3 slogan would be false: for $E_- = 0$ the spaces $\Lambda^2 E_-$ and $\Lambda^3 E_-$ are both zero, the contraction $C_\omega : 0 \rightarrow 0$ is trivially bijective, both operational predicates hold vacuously, but $\dim E_- = 0 \neq 3$. The transparency theorem `finrank_E_eq_zero_or_three` (v9) records the unconditional disjunction $\dim E_- = 0 \vee \dim E_- = 3$; the rank-3 conclusion is the case under `Nontrivial  $E_-$` . Both signatures are signature-locked in `AuditContract.lean`.

Stage D characterisation, both directions formalised (v10). For all four Yukawa stages the note now proves the full characterisation `Nonempty (Stage)  $\iff \dim E = 3$` . For Stage D, under `Nontrivial  $E$` :

$$\text{Nonempty } \{ Y : \text{YukawaTrilinearForm } K E \mid Y.\text{ContractionInjective} \wedge Y.\text{ContractionSurjective} \} \iff$$

The forward direction is `finrank_E_eq_three_of_condition`; the backward direction is `YukawaStageDExistence` proved via `Module.Basis.det` transported through `exteriorPower.alternatingMapLinearEquiv` and direct 3×3 matrix-determinant computation. The combined characterisation is `stage_D_iff_finrank_eq_3`.

What remains is upstream TFPT theory work. The Lean chain takes the Stage D quadruple $(\omega, \text{ContractionInjective}, \text{ContractionSurjective}, \text{Nontrivial } E_-)$ as input. The remaining open question is whether TFPT itself produces such a quadruple from the boundary kernel. Concretely:

- *Existence of ω .* Paper 2's Yukawa-type argument should produce an explicit alternating trilinear form on E_- . This is a representation-theoretic statement about the boundary kernel output, not a Lean engineering statement.
- *Verification of the two predicates.* Once ω is constructed, the two predicates need to be checked. These checks are local algebraic statements (injectivity: no nonzero v vanishes against all wedges; surjectivity: every dual functional on $\Lambda^2 E_-$ is realised as a contraction).

Neither task requires modifying the present Lean chain. They are TFPT mathematical work, properly upstream of this experiment.

Remaining limitations of the Lean chain itself.

- **Calderón: two structural layers formalised, analytic input of the polarisation open.** `CalderonProjector` proves $\pi^2 = \pi \Rightarrow \varepsilon^2 = 1$, and `BoundaryPolarization` (v10) shows that any direct-sum decomposition $H = H_+ \oplus H_-$ of a K -module produces such an idempotent via `Mathlib's IsCompl.projection`. What remains is the analytic Paper 1 theorem producing the boundary spectral decomposition itself — a question of boundary elliptic theory on Sobolev spaces, independent of the carrier algebra.
- **Higgs: algebraic Stage B and scheme-cohomology shadow formalised, Proj construction open.** `HiggsTopForm` identifies E_+ with the algebraic shadow of $H^0(\mathbb{P}^1, \mathcal{O}(1))$ (degree-1 binary forms), and v10's `HiggsSchemeCohomologyShadow` refines this further: the algebraic shadow is now exhibited as a `Mathlib-native` object, `MvPolynomial.homogeneousSubmodule` (Fin 2) K 1. What remains is the general algebraic-geometry theorem $H^0(\text{Proj } K[X, Y], \mathcal{O}(1)) \simeq K[X, Y]_1$, which lives in `Mathlib.AlgebraicGeometry.*` and is independent of TFPT.
- **Yukawa: Stage D both directions formalised.** v10's `YukawaStageDExistence` closes the backward direction: $\dim E_- = 3 \Rightarrow \exists \omega$ with `ContractionInjective` and `ContractionSurjective`, via `b.det` and direct 3×3 matrix-determinant computation. The characterisation $\exists \omega (CI \wedge CS) \iff \dim E_- = 3$ holds under `Nontrivial E_-`. What remains is the TFPT-internal question: does the boundary kernel produce *the same* ω (or one equivalent up to action of the carrier symmetries)?
- **Scope discipline.** The Lean chain formalises the $3 + 2$ carrier hypercharge. It does *not* formalise the full SM hypercharge spectrum on duals, exterior powers, and tensor products. The $\mathbb{Z}_6$ quotient and the derived-representation extensions are out of scope. See §10.

Other items deliberately out of scope.

- Full $\mathbb{Z}_6$ -quotient theorem for $G_{\text{phys}} = (SU(3) \times SU(2) \times U(1)_Y)/\mathbb{Z}_6$.
- Derived Standard-Model representations (dual, exterior powers, tensor products); the diagonal formalised here is the carrier hypercharge, not the full SM hypercharge spectrum.
- Analytic content of Paper 1 (Calderón polarisation, boundary elliptic theory).

8 How It Was Verified

The audit gate `scripts/audit.sh` enforces seven hard checks on every commit:

1. `lake build` succeeds (currently $\sim 1\,800$ jobs, ~ 10 s on top of the `Mathlib` cache).
2. No `sorry` or `admit` in the source.
3. No `axiom` or `constant` declarations.
4. No `unsafe`, `opaque`, or `partial` declarations.
5. `#print axioms` on every headline theorem returns a subset of `{propext, Classical.choice, Quot.sound}`.
6. Every headline name in `TfptCarrier/AuditCheck.lean` elaborates via Lean's `#check`. This catches renames, deletions, and elaboration failures that a text search would miss.
7. Every headline theorem in `TfptCarrier/AuditContract.lean` elaborates inside an example : `<exact intended type> := <theorem>` pattern. This is strictly stronger than `#check`: if a future refactor silently weakens a theorem's conclusion without renaming it, the example fails to typecheck and the audit fails.

Current verbatim output of `audit.sh`:

```

=== Build ===
[PASS] lake build succeeded

=== sorry / admit ===
[PASS] no sorry or admit found

=== axiom / constant declarations ===
[PASS] no axiom or constant declarations

=== unsafe / opaque / partial declarations ===
[PASS] no unsafe / opaque / partial declarations

=== axiom audit (#print axioms) ===
[PASS] all theorems use only [propext, Classical.choice, Quot.sound]

=== headline theorems (Lean #check) ===
[PASS] AuditCheck.lean elaborated; all headline #checks succeeded

=== theorem signatures (Lean example signature-lock) ===
[PASS] AuditContract.lean elaborated; all signature locks hold

=== Final verdict ===
AUDIT: PASS

```

```
#eval smoke tests in Sanity.lean:
```

```

info: TfptCarrier/Sanity.lean:27:0: 0
info: TfptCarrier/Sanity.lean:30:0:
  !![0, 0, 0, 0, 0; 0, 0, 0, 0, 0; 0, 0, 0, 0, 0;
    0, 0, 0, 0, 0; 0, 0, 0, 0, 0]

```

9 Changelog

v11 — review-pass closure and typed targets for the remaining upstream theorems (the present revision)

Relative to v10, this version is a closure-and-naming pass. It addresses the v10 reviewer's three actionable findings:

- **Editorial consistency.** The stale "Stage D characterisation as targeted future work" paragraph and the matching "Stage D backward" entry in §10, both inherited from v9, are removed: v10's YukawaStageDExistence closed that gap.
- **Wording.** "Upstream interfaces structurally closed" is reworded to "closed to the next algebraic layer", honestly reflecting that Calderón, Higgs Stage B, and Yukawa Stage D backward are now Lean-proved one-layer-up structural derivations, while the analytic / TFPT-internal content of *that next layer up* remains open.
- **Three-part claim contract.** The `tfptclaimcontract` environment is restructured to separate (a) direct `CarrierPremises` inputs, (b) Lean-proved upstream refinements, (c) open TFPT-origin theorems. The third group now lists three typed Lean targets for the open content.
- **New module** `BoundaryPolarization.ofCarrierInvolution`. The converse of v10's `BoundaryPolarization` given a carrier involution $\varepsilon^2 = 1$ on `Module.End K H`, the +1 and -1 eigenspaces form a `BoundaryPolarization`. This makes the Paper-1 interface a Lean-proved *algebraic equivalence*: the polarisation picture and the involution picture are two presentations of the same finite-dimensional carrier datum. *Not* the analytic Paper 1 derivation; the construction of the polarisation from boundary elliptic theory remains upstream.

- **New module** `SeamWindingInterface.lean`. Introduces `SeamWindingData` (m, n) , an abstract typed target for TFPT Paper 2's seam-class theorem " $[u_\Sigma] = 1$ produces a primitive oriented determinant-preserving carrier". Provides bidirectional bridges to and from `PrimitiveOrientedDeterminantCarrier`: the seam-winding interface is the existing rigidity input, renamed to make the upstream TFPT theorem visible.
- **New module** `BoundaryYukawaKernelInterface.lean`. Introduces `BoundaryYukawaKernel` $K E$, packaging the Yukawa Stage D quadruple $(\omega, CI, CS, \text{Nontrivial})$. This is *not* a new physical derivation; it is a typed renaming of the Stage D condition, supplied so that the open TFPT theorem "boundary kernel produces a canonical ω " has an explicit Lean-side conclusion. Forward direction `finrank_E_eq_three` reuses v10's Stage D discharge; backward direction of `FinrankEqThree` reuses v10's `YukawaStageDExistence` to give the characterisation `nonempty_iff_finrank_eq_three`. The *canonicity* of ω from the boundary kernel — whether the TFPT-internal construction produces the same form as the basis-built witness, up to carrier symmetries — remains the upstream physics question.
- **AuditCheck, AxiomCheck, AuditContract extended.** Fifteen new headline names added across the three new modules; all depend only on `{propext, Classical.choice, Quot.sound}`. `audit.sh` returns AUDIT: PASS.

After v11, the boundaries of the Lean-side carrier-rigidity chain are sharply visible. The three open TFPT-origin theorems each have a typed Lean target:

$$\begin{aligned} \text{Paper 1 analytic boundary theory} &\Rightarrow \text{BoundaryPolarization}, \\ \text{TFPT boundary kernel + reps} &\Rightarrow \text{BoundaryYukawaKernel}, \\ \text{seam class } [u_\Sigma] = 1 &\Rightarrow \text{SeamWindingData}. \end{aligned}$$

v10 — upstream interfaces closed to the next algebraic layer

Relative to v9, this version closes the three standing upstream gaps documented in the v9 limitations section. Each gap is replaced by a Mathlib-native structural Lean module:

- **New module** `BoundaryPolarization.lean`. Introduces the structure `BoundaryPolarization` $K H := \{H_{\text{plus}} H_{\text{minus}} : \text{Submodule } K H, \text{isCompl } H_{\text{plus}} H_{\text{minus}}\}$, the algebraic skeleton of the Paper 1 polarisation. The projection onto H_+ along H_- is an idempotent endomorphism (via Mathlib's `Submodule.IsCompl.projection` and its `IsIdempotentElem` lemma); a bridge to `CalderonProjector` feeds the `CalderonProjector` chain in `Module.End K H`, using the standard `Invertible (2 : End K M)` instance from Mathlib's `QuadraticForm.Basic`.
- **New module** `HiggsSchemeCohomologyShadow.lean`. Constructs a `LinearEquiv` between `DegreeOneTwoVarForm K` (the algebraic shadow used by `HiggsTopForm`) and Mathlib's `MvPolynomial.homogeneousSubmodule (Fin 2) K 1`, the degree-1 piece of the bivariate polynomial ring. The latter is the algebraic-geometry "Proj-side" presentation of $H^0(\mathbb{P}^1, \mathcal{O}(1))$. The proof uses `MvPolynomial.homogeneousSubmodule_one_eq_span_X` and `Submodule.mem_span_range_i` to extract coefficients; injectivity is by direct extraction of X_i -coefficients.
- **New module** `YukawaStageDExistence.lean`. Closes the Stage D backward direction: $\dim E = 3 \Rightarrow \exists \omega$ with `ContractionInjective` and `ContractionSurjective`. The construction uses Mathlib's `Module.Basis.det` (the determinant of vectors in a basis) transported via `exteriorPower.alternatingMapLinearEquiv` into a Yukawa form on $\Lambda^3 E$. Injectivity is proved by direct 3×3 matrix-determinant computation in the basis; surjectivity follows from $\dim E = \dim(\Lambda^2 E)^* = 3$ and Mathlib's `LinearMap.injective_iff_surjective_of_finrank_eq_fin`. The Stage D characterisation $\exists \omega (CI \wedge CS) \iff \dim E = 3$ now holds (under `Nontrivial E`).

- **AuditCheck, AxiomCheck, AuditContract extended.** Eighteen new headline names added; all depend only on $\{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$. `audit.sh` returns `AUDIT: PASS` with seven hard checks.

After v10, the conditional Lean theorem is

$$\begin{aligned} &\text{BoundaryPolarization } (H = H_+ \oplus H_-) \Rightarrow \pi^2 = \pi \Rightarrow \varepsilon^2 = \mathbf{1} \Rightarrow P_{\pm} \text{ idempotent} \\ &K[X_0, X_1]_1 \simeq \text{DegreeOneTwoVarForm} \Rightarrow \dim E_+ = 2 \\ &\dim E_- = 3 \text{ (with Nontrivial } E_-) \iff \exists \omega \text{ (CI} \wedge \text{CS)} \\ &m q_- + n q_+ = 0, q_- < 0 < q_+, \gcd = 1 \Rightarrow (q_-, q_+) = (-n/d, m/d) \end{aligned}$$

with every step a Lean theorem, no sorry, all axioms $\subseteq \{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$.

v9 — review-readiness pass and Stage D transparency

Relative to v8, this version is a consistency-and-transparency revision rather than a structural one. It addresses internal version drift identified in the v8 review and adds one unconditional transparency theorem at the Stage D interface:

- **New theorem** `YukawaTrilinearForm.finrank_E_eq_zero_or_three`. For any finite-dimensional K -vector space (no `Nontrivial` hypothesis) a trilinear form with `ContractionInjective` and `ContractionSurjective` satisfies $\dim E = 0 \vee \dim E = 3$. The two predicates are vacuously satisfied at $E = 0$, which is the mathematical reason the existing rank-3 theorem requires `Nontrivial E`.
- **Stage D slogans and statements carry `Nontrivial E` explicitly.** The frontbox, abstract, headline boxed equations, and interface table now state the rank-3 conclusion under `Nontrivial E`. The Lean theorem already carried this hypothesis; v9 fixes the documentation to match.
- **Predicate naming unified throughout the note.** All operational mentions use `ContractionInjective` and `ContractionSurjective`. The TFPT-side phrases "non-degenerate" and "primitive / no spectator" are retained only as docstring synonyms in mathematical prose.
- **Audit gate uniformly described as seven checks.** The earlier v6 wording "six checks" is removed; the `AuditContract.lean` signature-lock check is consistently the seventh.
- **Limitations section rewritten to match v8/v9.** The previous "Higgs Stage A only" and "Calderón interface is documentary only" wording is replaced by the accurate v9 status: Higgs has a Stage B algebraic shadow, Calderón has an algebraic idempotent-to-involution layer, and the remaining open content is precisely identified.
- **Calderón and Higgs wording sharpened.** The note no longer claims that Paper 1 or scheme cohomology is formalised. Calderón is presented as the algebraic step $\pi^2 = \pi \Rightarrow \varepsilon^2 = \mathbf{1}$; Higgs is presented as the algebraic shadow $E_+ \simeq \text{DegreeOneTwoVarForm } K$.
- **AxiomCheck, AuditCheck, AuditContract updated.** The three audit modules now include the new `finrank_E_eq_zero_or_three` headline. All audit checks continue to return `AUDIT: PASS`.

v8 — Calderón structural + Higgs Stage B

Relative to v7, this version upgrades two upstream interfaces from documentary wrappers to structural derivations, narrowing the "out of scope" column of the interface table:

- **New module** `CalderonProjector.lean`. Introduces the structure `CalderonProjector A := { $\pi : A, \pi^2 = \pi$ }`, the algebraic skeleton of the Paper 1 Calderón construction. Defines the spectral reflection $\varepsilon := 2\pi - 1$ and *proves* $\varepsilon^2 = 1$ via the elementary identity $(2\pi - 1)^2 = 4\pi^2 - 4\pi + 1 = 1$. The bridge `toCalderonCertificate` feeds the structural result into the existing `CalderonInterface.CalderonCertificate` wrapper, so the Paper 1 input is now derived from a single algebraic hypothesis: *the existence of an idempotent*.
- **New module** `HiggsTopForm.lean`. Introduces the structure `HiggsTopForm K Eplus := {Eplus  $\simeq$  DegreeOneTwoVarForm K}`, where `DegreeOneTwoVarForm K := {coeffX coeffY : K}` is the algebraic shadow of $H^0(\mathbb{P}_K^1, \mathcal{O}(1))$ — the space of homogeneous degree-1 polynomials $\alpha X + \beta Y$ in two formal variables over K . The Stage-B rank theorem $\dim E_+ = 2$ follows mechanically (`finrank_eq_two` on `DegreeOneTwoVarForm`, transported via the `HiggsTopForm` equivalence). A bridge `toHiggsIndexCertificate` produces the Stage A witness.
- **Updated audit gate**. `AxiomCheck.lean`, `AuditCheck.lean`, and `AuditContract.lean` extended with five new headline theorems. All five depend only on `{propext, Classical.choice, Quot.sound}`. `audit.sh` continues to return `AUDIT: PASS` with seven hard checks.

The Lean-side picture from v8 onwards is:

$$\begin{aligned} \text{Calderón projector } (\pi^2 = \pi) &\Rightarrow \varepsilon^2 = 1 \Rightarrow P_-, P_+ \text{ idempotent} \\ \text{Higgs Stage B } (E_+ \simeq \alpha X + \beta Y) &\Rightarrow \dim E_+ = 2 \\ \text{Yukawa Stage D } (\omega, \text{CI}, \text{CS}, \text{Nontrivial } E_-) &\Rightarrow \dim E_- = 3 \end{aligned}$$

with all three discharges now derived from *structural hypotheses*, not bare numerical / linear-equivalence assumptions.

Remaining Lean lemma gap. The Stage D characterisation backward direction — existence of a $(\omega, \text{ContractionInjective}, \text{ContractionSurjective})$ triple from $\dim E = 3$ — requires showing that the wedge map $E \rightarrow \text{Hom}(\Lambda^2 E, \Lambda^3 E)$, $v \mapsto (\eta \mapsto v \wedge \eta)$, is injective in dimension 3. The cleanest formalisation route extends a non-zero vector to a basis and uses `exteriorPower.iotaMulti_family_linearIndependent_ofBasis` to deduce $v \wedge e_1 \wedge e_2 \neq 0$. `Mathlib` does not yet have a direct lemma for the non-degeneracy of the wedge pairing $\Lambda^p E \otimes \Lambda^{n-p} E \rightarrow \Lambda^n E$, so this is a single missing infrastructure lemma; the remainder of the proof is bookkeeping.

v7 — Signature lock, predicate rename, interface discipline

Relative to v6, this version hardens the audit interface, neutralises a semantically-loaded naming choice, and tightens the editorial framing around what the note does and does not claim:

- **New** `AuditContract.lean` **module**. Each headline theorem is now used in an `example : <exact intended type> := <theorem>` pattern. This is strictly stronger than the `#check` audit of `AuditCheck.lean`: it detects silent weakening of theorem statements that `#check` would miss (since a weakened theorem still elaborates under its own name). The `audit.sh` gate now has *seven* checks; the new seventh requires `AuditContract.lean` to elaborate.
- **Stage D predicate rename**. `YukawaTrilinearForm.Nondegenerate` $\rightarrow$ `ContractionInjective` and `YukawaTrilinearForm.Primitive` $\rightarrow$ `ContractionSurjective`. The new names are purely operational — they describe the bijectivity of C_ω — and leave the semantic content

("non-degenerate", "primitive") to the docstrings. A combined predicate `PrimitiveIndecomposableYukawaC` is available for the TFPT-side phrase. This removes the "primitive renamed to surjective" attack surface.

- **New `OrientedDeterminantCarrier.lean` module.** The three-part TFPT carrier condition — trace-zero, orientation, primitivity — is now packaged as a single Lean structure `PrimitiveOrientedDeterminantCarrier (m n : ℕ)`. Producing such a structure is the formal target for a future TFPT-internal theorem "seam winding $[u_\Sigma] = 1$ implies a primitive oriented determinant-preserving carrier".
- **Interface table.** A new §6 maps every load-bearing TFPT Paper 2 claim to its current Lean status and the missing upstream proof. This separates "formalised", "interfaced", and "out of scope" without ambiguity.
- **Claim Contract refinement.** The contract now distinguishes *direct* inputs to `CarrierPremises` (the rank witnesses and the boundary involution) from *refinable* upstream interfaces (Stages B/C/D feeding the Yukawa rank witness).
- **Editorial fixes.** The running header is overridden for this note (it previously inherited the TFPT 4.5 series header). The "v3" relic in the Result section is corrected.

v6 — Yukawa Stage D and Calderón interface

Relative to v5, this version adds two new modules and resolves the standing review-level critique that "Stage C is data, not derivation":

- Added module `YukawaTrilinearForm.lean` with the structure `YukawaTrilinearForm K E := {ω : Λ3E →ℓK K}` and two Prop predicates (originally named `Nondegenerate` and `Primitive`, renamed in v7 to `ContractionInjective` and `ContractionSurjective` respectively).
- Constructed the contraction map `contraction : E →ℓK ((Λ2E) →ℓK K)` from ω explicitly using `exteriorPower.alternatingMapLinearEquiv` together with `AlternatingMap.curryLeft`. The contraction is now a Lean *definition*, not a hypothesis.
- Proved `contractionEquivOfBoth`: under both predicates the contraction upgrades to a linear isomorphism. Hence `toPrimitiveIndecomposableYukawaCoupling`: Stage D induces Stage C as a theorem.
- Proved `finrank_E_eq_three`: Stage D directly implies $\dim E = 3$ through the full tower.
- Added module `CalderonInterface.lean` with the structure `CalderonCertificate` that wraps Paper 1's Calderón-induced carrier involution as a Lean datum and feeds it into the existing `CarrierInvolution` pipeline. The Paper 1 input is now an explicit interface, not a silent consumption.
- Updated `AxiomCheck.lean` and `AuditCheck.lean` with eight new theorem names. All eight depend only on `{propext, Classical.choice, Quot.sound}`. `audit.sh` returns `AUDIT: PASS`.

The Yukawa interface is now a four-stage tower (under `Nontrivial E_` at the Stage D entry point)

$$\underbrace{(\omega, \text{CI}, \text{CS})}_{\text{Stage D}} \Rightarrow \underbrace{C_\omega : E_- \simeq (\Lambda^2 E_-)^*}_{\text{Stage C}} \Rightarrow \underbrace{\Lambda^3 E_- \simeq K}_{\text{Stage B}} \Rightarrow \underbrace{E_- \simeq K^3}_{\text{Stage A}} \Rightarrow \dim E_- = 3.$$

Every step is a Lean theorem. The primary algebraic content sits at Stage D: an alternating trilinear form plus two explicit structural predicates.

v5 — Yukawa Stage C

Relative to v4, this version closes the Yukawa interface *within* Lean by introducing a third stage:

- Added module `YukawaPrimitive.lean` with the structure `PrimitiveIndecomposableYukawaCoupling` $K\ E := \{\text{contraction} : E \simeq_{\ell K} (\Lambda^2 E)^*, \text{nontrivial} : \text{Nontrivial } E\}$.
- Proved `finrank_E_eq_three` via the dimension chain $\dim E \stackrel{\text{iso}}{=} \dim(\Lambda^2 E)^* \stackrel{\text{dual}}{=} \dim \Lambda^2 E \stackrel{\text{ext. pow.}}{=} \binom{\dim E}{2}$ combined with the elementary fact that $n = n(n-1)/2$ and $n \geq 1$ force $n = 3$.
- Proved the converse of `FinrankEqThree` and the characterisation `nonempty_iff_finrank_eq_three`.
- Bridged Stage C to Stage B via `toYukawaTopForm`, and transitively to Stage A via `toYukawaRankCertificate`.
- Updated `AxiomCheck.lean` and `AuditCheck.lean` with the five new theorem names. All five depend only on `{propext, Classical.choice, Quot.sound}`.

The Yukawa interface is now structured as a three-stage tower

$$\underbrace{C_\omega : E_- \simeq (\Lambda^2 E_-)^*}_{\text{Stage C}} \Rightarrow \underbrace{\Lambda^3 E_- \simeq K}_{\text{Stage B}} \Rightarrow \underbrace{E_- \simeq K^3}_{\text{Stage A}} \Rightarrow \dim E_- = 3,$$

with each step a Lean theorem and a characterisation establishing the equivalence to $\dim E_- = 3$ at every stage. The *conceptual* input to Stage C is a contraction iso of E_- with $(\Lambda^2 E_-)^*$, which is the mathematical core of "primitive indecomposable Yukawa type" with no physics words.

v4 — Yukawa Stage B

In v4 the Yukawa rank discharge was promoted from a bare Stage-A rank witness ($E_- \simeq K^3$, a numerical assumption in linear-equivalence dress) to a Stage-B algebraic hypothesis ($\Lambda^3 E_- \simeq K$, a structural property of E_-).

- Added module `YukawaTopForm.lean` with the structure `YukawaTopForm` $K\ E := \{\text{topFormEquiv} : \Lambda^3 E \simeq_{\ell K} K\}$.
- Proved `YukawaTopForm.finrank_E_eq_three` using Mathlib's `exteriorPower.finrank_eq` and `Nat.choose_eq_one_iff`.
- Proved the converse of `FinrankEqThree` and the characterisation `nonempty_iff_finrank_eq_three`.
- Bridged Stage B to Stage A via `toYukawaRankCertificate`.
- Updated `AxiomCheck.lean` and `AuditCheck.lean` with the five new theorem names. All five depend only on `{propext, Classical.choice, Quot.sound}`.

After v4, the conceptual gap shrunk to

$$\text{"primitive indecomposable Yukawa type"} \implies \text{Nonempty } (\text{YukawaTopForm } K\ E_-).$$

v5 closes this gap with the Stage C contraction iso definition.

v3 — correcting overclaims

Relative to v2, v3 corrected three classes of overclaim:

1. **Scope of the title and the abstract.** "SM Hypercharge Spectrum" $\rightarrow$ "3 + 2 Hypercharge Carrier". The note no longer suggests it has formalised all SM chiral- multiplet hypercharges; only the carrier multiplet $\text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$.
2. **Status of the Stage-A certificates.** "Not assumptions" $\rightarrow$ "structured Stage-A hypotheses, not bare numerical assumptions". The Higgs and Yukawa certificates are hypotheses in Lean. Calling them not-assumptions misrepresents the type-theoretic status. The Yukawa structure is renamed from `PrimitiveYukawaTypeCertificate` to `YukawaRankCertificate`, accurately describing what it actually carries.
3. **Logical direction of the dependency graph.** The v2 graph had three incorrect arrows. v3 routes `DeterminantCharacter` $\rightarrow$ `LatticeRigidityGeneral` (not the reverse); removes the spurious `HiggsIndexCertificate` $\rightarrow$ `YukawaRankCertificate` edge; and removes the `InvolutionProjectors` $\rightarrow$ `LatticeRigidityGeneral` edge.

Two further substantive additions:

- **Bundled main theorems strengthened.** v3 adds `CarrierPremises.primitive_pair_eq_sm`, `CarrierPremises.carrier_polynomial_Y`, and the top-level `hypercharge_carrier_packet` bundling all three conclusions $(q_-, q_+) = (-2, 3)$, $\text{Tr } Y = 0$, $6Y^2 - Y - \mathbf{1} = \mathbf{0}$. The previous version only bundled $\text{Tr } Y = 0$, which was weaker than the abstract claimed.
- **Universal- λ determinant character.** v3 adds `trace_zero_of_det_one_forall_rat` as the proper headline, with `trace_zero_of_det_one_at_two` kept as a single-point helper.

The v2-level claim now reduces to its accurate form, refined for v9:

Lean verifies the algebraic carrier graph from:

- (i) a Calderón *idempotent* (`CalderonProjector`: $\pi^2 = \pi \Rightarrow \varepsilon^2 = 1$),
 - (ii) a Higgs Stage-B *polynomial structure* ($E_+ \simeq \alpha X + \beta Y \Rightarrow \dim E_+ = 2$),
 - (iii) a Yukawa Stage-D *trilinear form* ($\omega : \Lambda^3 E_- \rightarrow K$ with `ContractionInjective`, `ContractionSurjective`, and `Nontrivial` $E_- \Rightarrow \dim E_- = 3$),
 - (iv) a primitive oriented determinant-preserving carrier condition,
- down to the 3 + 2 hypercharge carrier $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$
with $(q_-, q_+) = (-2, 3)$, $\text{Tr } Y = 0$, and $6Y^2 - Y - \mathbf{1} = \mathbf{0}$.

All three rank/involution discharges are now *structural*: the Calderón input is an algebraic idempotent (not a wrapped involution), the Higgs input is a polynomial-space iso (not a bare rank witness), and the Yukawa input is an alternating trilinear form plus two operational predicates (not a packaged iso).

10 Next Steps

In rough order of ascending intellectual difficulty:

1. **CI workflow.** Wire `audit.sh` into a GitHub Actions workflow that refuses any merge with `AUDIT: FAIL`.

2. **Higgs certificate from $\mathbb{P}^1$ cohomology.** Build a Lean construction of `HiggsIndexCertificate` from the cohomology of $\mathcal{O}(1)$ on $\mathbb{P}^1$.
3. **Higgs scheme-cohomology realisation.** The Stage B `HiggsTopForm` now identifies E_+ with the algebraic shadow of $H^0(\mathbb{P}^1, \mathcal{O}(1))$; a future module should build the scheme-theoretic $H^0(\mathbb{P}_K^1, \mathcal{O}(1))$ and exhibit an iso to `DegreeOneTwoVarForm K`. This is Mathlib algebraic geometry work and reasonably sized.
4. **Calderón boundary spectral pair.** The Stage A+ `CalderonProjector` derives $\varepsilon^2 = 1$ from an algebraic idempotent, and `v10's BoundaryPolarization` derives the idempotent from a complement-pair $H = H_+ \oplus H_-$. What remains is the analytic Paper-1 theorem producing the polarisation itself from boundary elliptic theory (Calderón operator on Sobolev spaces). This is a major independent formalisation project.
5. **Yukawa: existence of ω from TFPT canonically.** The Lean chain now provides full existence: $\dim E_- = 3 \Rightarrow \exists \omega$ with `ContractionInjective` and `ContractionSurjective` (`YukawaStageDExistence, v10`). What remains is the TFPT-internal question: does the boundary kernel produce *the same canonical* ω (or one equivalent up to the action of the carrier symmetries)? The abstract bridge `BoundaryYukawaKernelInterface` (`v11`) gives a Lean-typed target for this future TFPT theorem.
6. **Derived SM representations.** Implement `LinearMap.dualHypercharge`, `ExteriorPower.hypercharge`, `TensorProduct.hypercharge`, so that the project can claim "SM hypercharge spectrum" rather than "carrier hypercharge".
7. **$\mathbb{Z}_6$ -quotient theorem.** The kernel of the carrier action of $SU(3) \times SU(2) \times U(1)$ on $\mathbb{K}^5$ as $\mathbb{Z}_6$, using `Mathlib.LinearAlgebra.UnitaryGroup`.
8. **Mathlib upstream.** After (2) and (4), `Polarization`, `InvolutionProjectors.toPolarization`, and the Mathlib bridge make a clean PR candidate.

11 Result

Exported objects

Lean-verified outputs (v11).

- Abstract carrier polynomial: `Polarization.sixY_carrier_polynomial`.
- Boundary involution $\Rightarrow$ projectors: `InvolutionProjectors.CarrierInvolution.toPolarization`.
- Mathlib bridge: `MathlibBridge.Polarization.toCompleteOrthogonalIdempotents`.
- General integer rigidity: `LatticeRigidityGeneral.primitive_trace_free_pair_general`.
- Structural trace identity: `TraceProjection.trace_linear_combination_of_idempotents`.
- Determinant character, universal- λ headline: `DeterminantCharacter.trace_zero_of_det_one_forall_1`.
- Higgs Stage-A discharge: `HiggsIndexShadow.HiggsIndexCertificate.finrank_Eplus_eq_two`.
- Yukawa Stage-A discharge (rank witness): `YukawaRank.YukawaRankCertificate.finrank_Eminus_eq_th`.
- Yukawa Stage-B discharge (algebraic top form): `YukawaTopForm.YukawaTopForm.finrank_E_eq_three`.
- Yukawa Stage-B characterisation: `YukawaTopForm.YukawaTopForm.nonempty_iff_finrank_eq_three`.
- Stage-B $\rightarrow$ Stage-A bridge: `YukawaTopForm.YukawaTopForm.toYukawaRankCertificate`.
- Yukawa Stage-C discharge (contraction iso): `YukawaPrimitive.PrimitiveIndecomposableYukawaCoupli`.

- Yukawa Stage-C characterisation: `YukawaPrimitive.PrimitiveIndecomposableYukawaCoupling.nonempty`
- Stage-C $\rightarrow$ Stage-B bridge: `YukawaPrimitive.PrimitiveIndecomposableYukawaCoupling.toYukawaTopForm`
- Yukawa Stage-D primary datum and constructed contraction: `YukawaTrilinearForm.YukawaTrilinearForm.contraction`.
- Stage-D operational predicates: `YukawaTrilinearForm.YukawaTrilinearForm.ContractionInjective` and `.ContractionSurjective`.
- Stage-D combined predicate (`PrimitiveIndecomposableYukawaCondition`) and rank-3 discharge under Nontrivial E : `YukawaTrilinearForm.YukawaTrilinearForm.finrank_E_eq_three` and `.finrank_E_eq_three_of_condition`.
- Stage-D transparency form (no Nontrivial, $\dim \in \{0,3\}$): `YukawaTrilinearForm.YukawaTrilinearForm.finrank_E_eq_zero_or_three`.
- Stage-D $\rightarrow$ Stage-C bridge: `YukawaTrilinearForm.YukawaTrilinearForm.toPrimitiveIndecomposable`
- Calderón certificate interface: `CalderonInterface.CalderonCertificate.toCarrierInvolution`.
- Calderón structural derivation (idempotent $\Rightarrow$ involution): `CalderonProjector.CalderonProjector.eps_sq`.
- Calderón upstream (v10): boundary polarisation $\Rightarrow$ idempotent: `BoundaryPolarization.BoundaryPolarization.projector_isIdempotent` and `toCalderonProjector`.
- Higgs Stage B discharge (polynomial space $\Rightarrow \dim E_+ = 2$): `HiggsTopForm.HiggsTopForm.finrank_Eplus_eq_two`.
- Higgs scheme-cohomology shadow (v10): `HiggsSchemeCohomologyShadow.toDegreeOneTwoVarForm` (LinearEquiv between the algebraic shadow and `MvPolynomial.homogeneousSubmodule (Fin 2) K 1`).
- Yukawa Stage D backward (v10): `YukawaStageDExistence.exists_yukawaTrilinearForm_of_finrank_eq_three` and the characterisation `stage_D_iff_finrank_eq_three` (under Nontrivial E).
- Boundary polarisation converse (v11): `BoundaryPolarization.ofCarrierInvolution` — the Paper-1 interface is a Lean-proved *algebraic* equivalence between the polarisation picture and the involution picture (two presentations of the same finite-dimensional carrier datum; not the analytic construction).
- Seam-winding interface (v11): `SeamWindingInterface.SeamWindingData.toPrimitiveOrientedDeterminant` and the converse, with the SM specialisation `sm_pair_eq`.
- Boundary Yukawa kernel interface (v11): `BoundaryYukawaKernelInterface.BoundaryYukawaKernel.finrank_eq_three_ofFinrankEqThree`, and the characterisation `nonempty_iff_finrank_eq_three`. *Typed packaging of the Stage D condition — not a new physical derivation. The canonicity of ω from the TFPT boundary kernel remains upstream physics work.*
- Bundled main theorems: `CarrierPremises.primitive_pair_eq_sm`, `CarrierPremises.trace_Y_eq_zero`, `CarrierPremises.carrier_polynomial_Y`, `CarrierPremises.hypercharge_carrier_packet`.
- Concrete carrier polynomial: `Hypercharge.Y_carrier_polynomial`.

Audit outputs. 1788 build jobs green; zero sorry; zero admit; zero unsafe / opaque / partial; zero domain-specific axioms; every headline theorem depends only on `{propxt, Classical.choice, Quot.sound}`.

A Reproducing the Build and Audit

```
# 1. Install elan (Lean toolchain manager)
curl -sSf https://raw.githubusercontent.com/leanprover/elan/master/elan-init.sh \
  | sh -s -- -y --default-toolchain none
export PATH="$HOME/.elan/bin:$PATH"

# 2. Enter the project
cd experiments/lean4-carrier-rigidity

# 3. Fetch Mathlib and its pre-built oleans
lake update # ~30 s
lake exe cache get # ~3 min, ~1 GB

# 4. Build the project
lake build # ~8 s on top of cache; 1788 jobs

# 5. Run the seven-check audit gate
./scripts/audit.sh
```

Expected trailer:

```
=== Final verdict ===
AUDIT: PASS
```